



KRMU/SOET/09

PROPOSED LAB EXPERIMENTS

Fundamental C++ for Robotics

Department:	School of Engineering & Technology	
Program:	B.Tech (Robotics)	
Course Name:	Fundamental C++ for Robotics	
Course Code	L-T-P	Credits
ETCCCPL205	3-0-2	4

Experiment Mapping Table:

Experiment No.	Experiment Title	Mapped CO/COs
1	Installation and Configuration of GCC/G++, VS Code, and GDB with Development of a First Robotics Initialization Program in C++	CO1
2	Design and Implementation of an Obstacle Detection Decision System using if-else Control Logic in C++	CO1
3	Simulation of Repetitive Robotic Movement using for, while, and do-while Loop Constructs	CO1
4	Development of Modular Robotic Action Functions for Movement and Direction Control in C++	CO1
5	Design of a Menu-Driven Robot Control Application Integrating Functions, Loops, and switch-case Statements	CO1
6	Analysis of Stack, Static, and Heap Memory Allocation through a C++ Memory Layout Visualization Program	CO2
7	Implementation of Dynamic Sensor Data Storage and Processing using Pointers and Dynamic Memory Allocation in C++	CO2
8	Development of a Robot Position Tracking System using STL Vector, Iterators, and Range-Based Loops	CO2

Experiment No.	Experiment Title	Mapped CO/COs
9	Design and Simulation of a Robot Command Buffer using STL Queue and FIFO-Based Execution	CO2
10	Implementation of Safe Sensor Input Validation using Exception Handling and Runtime Error Management in C++	CO2
11	Design of a Robotic Actuator Control Class using Object-Oriented Programming Principles and Encapsulation	CO3
12	Implementation of Constructors and Destructors for Safe Lifecycle and Resource Management in Robotic Objects	CO3
13	Development of a Robot Class Hierarchy using Inheritance and Runtime Polymorphism for Multiple Robot Types	CO3
14	Design of an Integrated Robot System using Composition-Based Modeling of Sensors and Controllers	CO3
15	Implementation of Safe Ownership and Automatic Resource Cleanup in Robotic Systems using Smart Pointers	CO3
16	Design and Implementation of Embedded-Style Robot Status Management using Bitwise Flags and Enumerations	CO4, CO5
17	Measurement and Profiling of Sensor Function Execution Time using chrono for Latency Analysis in Robotics Applications	CO4, CO5
18	Implementation of Sensor Data Serialization and Deserialization using C++ Structures and Byte-Level Data Packing	CO4, CO5
19	Development and Timing Enforcement of a Periodic Real-Time Control Loop for Simulated Robotic Operations	CO4, CO5
20	Evaluation of Deadline Satisfaction, Jitter, and Safety Constraints in a Real-Time Robotic Control Loop through Timing Analysis Report	CO4, CO5

Prepared in editable format for academic use.